

Peer Review Information

Journal: Nature Human Behaviour

Manuscript Title: Evidence for Early Dispersal of Domestic Sheep into Central Asia

Corresponding author name(s): William T.T. Taylor & Svetlana Shnaider

Editorial Notes:

Reviewer Comments & Decisions:

Decision Letter, initial version:

19th November 2020

Dear Dr Taylor,

Thank you once again for your manuscript, entitled "Evidence for Early Dispersal of Domestic Sheep into Central Asia: genetic, archaeozoological, and biomolecular insights from Obishir V," and for your patience during the peer review process.

Your manuscript has now been evaluated by 3 reviewers, whose comments are included at the end of this letter. Although the reviewers find your work to be of interest, they also raise some important concerns. We are very interested in the possibility of publishing your study in Nature Human Behaviour, but would like to consider your response to these concerns in the form of a revised manuscript before we make a decision on publication.

To guide the scope of the revisions, the editors discuss the referee reports in detail within the team, including with the chief editor, with a view to (1) identifying key priorities that should be addressed in revision and (2) overruling referee requests that are deemed beyond the scope of the current study. We hope that you will find the prioritised set of referee points to be useful when revising your study. Please do not hesitate to get in touch if you would like to discuss these issues further.

- 1) Please revise the manuscript to explain clearly the novelty and advance that this work represents, in response to Reviewer 3's concerns;
- 2) In response to Reviewer 3's queries regarding your analytical approach, please update your methods accordingly;
- 3) Please respond in full to all other reviewer comments.

Finally, your revised manuscript must comply fully with our editorial policies and formatting requirements. Failure to do so will result in your manuscript being returned to you, which will delay its consideration. To assist you in this process, I have attached a checklist that lists all of our requirements. I have also attached a template manuscript file that exemplifies our policies and formatting requirements. If you have any questions about any of our policies or formatting, please don't hesitate to contact me.

In sum, we invite you to revise your manuscript taking into account all reviewer and editor comments. We are committed to providing a fair and constructive peer-review process. Do not hesitate to contact us if there are specific requests from the reviewers that you believe are technically impossible or unlikely to yield a meaningful outcome.

We hope to receive your revised manuscript within four to eight weeks. We understand that the COVID-19 pandemic is causing significant disruption for many of our authors and reviewers. If you cannot send your revised manuscript within this time, please let us know - we will be happy to extend the submission date to enable you to complete your work on the revision.

With your revision, please:

- Include a "Response to the editors and reviewers" document detailing, point-by-point, how you addressed each editor and referee comment. If no action was taken to address a point, you must provide a compelling argument. This response will be used by the editors to evaluate your revision and sent back to the reviewers along with the revised manuscript.
- Highlight all changes made to your manuscript or provide us with a version that tracks changes.

Please use the link below to submit your revised manuscript and related files:

[REDACTED]

Note: This URL links to your confidential home page and associated information about manuscripts you may have submitted, or that you are reviewing for us. If you wish to forward this email to co-authors, please delete the link to your homepage.

We look forward to seeing the revised manuscript and thank you for the opportunity to review your work. Please do not hesitate to contact me if you have any questions or would like to discuss these revisions further.

Sincerely,

Charlotte Payne
Editor
Nature Human Behaviour

Reviewer expertise:

Reviewer #1: aDNA; archaeozoology of domestic livestock; cementum analysis of teeth

Reviewer #2: aDNA; archaeozoology of domestic livestock

Reviewer #3: aDNA; archaeozoology of domestic livestock

REVIEWER COMMENTS:

Reviewer #1:

Remarks to the Author:

I recommend publication with very minimal edits. There are some small formatting issues that I'm sure will be resolved with typesetting (like centered figure captions). Overall most of the manuscript was relatively concise and clear. I did find the use of "these two taxa" in the Discussion a little opaque; are the authors referring to wild or domestic forms when they say "a ratio of roughly 3:1 between these two taxa have characterized pastoralist archaeological assemblages in Central and Inner Asia since the Bronze Age"; I think that those few sentences could be clarified a bit. Otherwise I think the amount of data provided in the supplemental information is appropriate and there is a good balance between what is presented and discussed in the main article and what is supplied for the reader. Though it is a small sample size, obviously the data are there to support the authors' argument that there were domestic sheep present at this early date, and that is a very important finding that should be published in this journal. I think the paper lays a foundation for future research in the region and makes a substantial contribution to our understanding of the spread of domesticates eastwards of the Fertile Crescent.

Reviewer #2:

Remarks to the Author:

The article "Evidence for Early Dispersal of Domestic Sheep into Central Asia: genetic, archaeozoological, and biomolecular insights from Obishir V" by Taylor et al. represents a scientific study of extreme importance in the field of zooarchaeology, covering the ancient genetics of domestic sheep, dispersal of domestic sheep after its initial domestication, and earliest insights into animal husbandry in Central Asia – topics that are long debated and are still debated by the scholars in different fields. Although the study relies on rather scanty material from the site of Obishir V, Kyrgyzstan, the applied methods of interdisciplinary nature definitely provide a solid argument for a very important discovery: pastoral livestock species reached Central Asia millennia earlier as previously recognised, by around 6000 BCE. These results are a valuable input to the studies of domestication, dispersal of the domestic species from the domestication centre, and human/cultural migrations.

The article text is very well written and enjoyable to read. The figures, tables, and supplementary appendices illustrate and support the main text sufficiently. The selected methods have been well

described, applied, and justified. Please note that the reviewer has no experience in thermoluminescence dating and cementum analysis and has somewhat experience in genetics, meaning that any potential shortages in these sections could have been missed.

There are only a few very minor remarks I would like to point out to the authors and when addressed, these would hopefully improve the quality of the article even further. These remarks are purely from the editorial point of view and do not diminish the quality of the article.

1. In Introduction, paragraphs 3 and 4, you have mentioned that “Domesticated sheep and goat are also thought to have been an important part of the subsistence economy at Djeitun (33 - 36).” and “Importantly, scholars have also argued that the Hissar economy included goat and sheep pastoralism (37).”. If possible, maybe you could add to those sentences, on which evidence these earlier suggestions rely – have the previous scholars relied here on the presence of ovicaprid remains?

2. In Discussion, paragraph 1, you have written that “Although wild Ovis and Capra in this region include the large argali sheep (*Ovis ammon*) and Siberian ibex (*Capra sibirica*), a ratio of roughly 3:1 between these two taxa have characterized pastoralist archaeological assemblages in Central and Inner Asia since the Bronze Age (22, 52), and are also typical of other early western Eurasian pastoral assemblages (53). Based on the results of ZooMS, these two taxa appear to comprise the majority of the unidentifiable fragments in the Layer 2 faunal assemblage.”. Please clarify, which “two taxa” you have meant in the last sentence? I believe it should be *Ovis* sp. and *Capra* sp.?

3. In Discussion, under subchapter Evidence for pastoral management, paragraph 2, I believe there is something gone missing in the sentence “Our new findings indicate that the subsistence behaviors that underpinned these nascent interaction”.

4. Few missing full stops and spaces noticed both in the main text and Supplementary Appendix.

5. Regarding the figures, I have few comments:

- a. Few of them appear to be of low resolution. However, if this is due to the submission process and better-quality figures will be uploaded after acceptance, please disregard this comment.
- b. In Figure 3 (main text) and Figure C1 in Supplementary Appendix, the text is too small to follow – I would suggest enlarging it a bit.
- c. In Figure 3A (main text) I would like to see NISP values for each bar – it would make following the material details much easier.
- d. In Figure 3B (main text), I can see groupings for cervidae/bovidae (4) and deer (5), but these are not listed in the text as the other groupings – maybe to list them in the text as well?
- e. Figure 5 is a single figure, but in the main text it has been referred to as Figures 5A and 5B.

6. Regarding the Supplementary Appendix, I have few comments:

- a. Consider improving the quality of Table B2.
- b. Large tables like Table C1, Table C2, Table F2, and Table F3 should be formatted for better readability or as written in the journal guidelines, “Tables larger than one page should be provided as tabular data files than a PDF.” In this case, just formatting them in the already existing Word file could work as well. Additionally, Table C2 is currently in both portrait and landscape.
- c. In Table F1 caption, you mention numbering for sheep, but I can only see *Capra* in the table.

Reviewer #3:

None

Author Rebuttal to Initial comments
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9 December, 2020

Dear Dr. Payne,

Attached, please find a revised version of our manuscript, “Evidence for Early Dispersal of Domestic Sheep into Central Asia: genetic, archaeozoological, and biomolecular insights from Obishir V” for your consideration. The paper has been revised based on suggestions from three reviewers, who all found the manuscript to be well-written, with well-supported conclusions of broad significance for our understanding of human prehistory. In its revised form, we have addressed a number of minor issues with the text and supplementary material identified by the reviewers, and I hope you will find it to be a valuable contribution to *Nature: Human Behavior*.

Below, you will find a point-by-point description of the minor revisions we made in response to these reviewer comments, as well as a few minor points of clarification.

Reviewer 1

Reviewer 1 requested that we center our figure captions, we have done so.

Reviewer 1 also identified ambiguity in reference to “these two taxa” in the first paragraph of the discussion, an issue that was also highlighted by Reviewer 2. As Reviewer 2 had correctly inferred, our intent here was to refer to the genus *Ovis* and *Capra*. We have clarified the wording to this section to make this point clear.

Reviewer 2

In comments that were echoed by Reviewer 3, Reviewer 2 requested clarification on the evidence used to infer the presence of domestic animals at Djeitun and in sites of the Hissar culture (Introduction, paragraphs 3 and 4). In our revised manuscript, we have added several sentences establishing that only indirect measures – a high frequency of *Ovis* and *Capra* remains that appear smaller than the region’s wild taxa – were used by earlier scholars to generate this hypothesis. Our revised introduction now makes it clear that the suggestion of domestic sheep and goat at Djeitun has not been rejected, but neither has it been verified through scientific methods. We have also updated a sentence in Paragraph 5 to clarify that no similar arguments have been made for the Ferghana region, which lies nearly 1400 km to the northeast of Djeitun. The revised language and new sentences in this section now make it clear that our study, while contextually supported by earlier work in Turkmenistan and Tajikistan hypothesizing an early eastward dispersal of Neolithic economies, is testing and confirming a new, previously untested hypothesis.

Main Text

- Reviewer 2 identified a sentence that was inadvertently deleted in the Discussion/Evidence for pastoral management subsection, paragraph 2. We have restored this sentence.
- We have proofed the main text and supplementary material for missing punctuation, extra spaces, and missing spaces, per Reviewer 2’s request.

Figures

- Reviewer 2 noted that the figures in the review copy were low-resolution – we suspect that this is related to the submission process, as our uploaded figures were indeed high resolution.
- For Figure 3 and Figure C1 in the Supplementary Appendix, we have increased the size of label text to increase legibility.
- We added NISP values to each bar in Figure 3A.
- In reference to Figure 3B, we have added text descriptions to the groupings for Cervidae/Bovidae and deer/saiga/sheep/gazelle, as requested by Reviewer 2.
- We have corrected references to Figure 5 in the main text, to correctly refer to Figure 5 and Supplementary Figure F1 (instead of 5A and 5B, as incorrectly noted in the first manuscript).

Supplementary Appendix

- Table B2, which was incorrectly included as an embedded image file rather than as text in the initial submission, has been revised to restore the text and thereby improve quality.
- Caption reference to sheep in Table F1 removed, to retain only information to *Capra* sp.
- Tables C1, C2, F2, and F3 have been formatted as tabular data, and included in a separate “Supplementary Data” file, correcting the formatting and readability issues identified by Reviewer 2. Appropriate references in the text have been updated to reflect this change.

Reviewer 3

This reviewer’s comments were also largely positive, with the commenter highlighting the study’s convincing and significant results. In addition to the comments addressed above, Reviewer 3 also requested a number of minor revisions which we have addressed here.

- Reviewer 3 requested that we add page and line numbers to the manuscript. We have done so.
- Reviewer 3 requested that we add information about how the tree in Figure 6 was built to the text. We have added this information to Supplementary Appendix F, supported with a note to this effect to the caption for Figure 6 in the main text.

Reviewer 3 found our mtDNA analysis to be brief but well-supported, but inquired as to the reasoning behind our selection of Maximum Parsimony over other alternatives (such as Maximum Likelihood with bootstrapping, neighbor-joining with bootstrapping, or Bayesian

methods). We respond that Maximum Parsimony still has currency in some areas of ancient genomics, as it minimizes the number of evolutionary events needed to explain the data. In

comparison, Maximum Likelihood and Bayesian inference methods incorporate an evolutionary model and prior knowledge about the data, respectively, that may not be as suited to all circumstances. However, we have repeated our analyses using a Maximum Likelihood approach with bootstrapping, and added this information to Supplementary Appendix F. The Maximum Likelihood tree supports the typology of the Maximum parsimony tree proposed in the main text.

Regarding the PCA, we based our initial assertion that Obishir sheep fell within the variation of *O. aries* on our phylogenetic analyses, which included important wild sheep taxa including the argali sheep, *O. ammon*, the Urial sheep, *O. vignei*, and the Cypriot mouflon, *O. orientalis ophion*. However, we recognize the added value of incorporating wild sheep data into our subsequent Principal Components Analysis. While whole-genome data for wild sheep taxa are less widely available, we were able to isolate a number of Asiatic mouflon genotypes –

the most closely related wild sheep taxa to the domestic *O. aries* – from a recently published study (<https://www.nature.com/articles/s41467-020-16485-1>). In our revised analysis, we have incorporated these wild sheep genomes into our PCA along with two other wild taxa (*O. canadensis* and *O. dalli*), which support our interpretation that the specimens are indeed domestic sheep.

Finally, Reviewer 3 also remarked that Table 3 was lacking information about postmortem damage patterns, and suggested that we add a supplementary MapDamage plot to support this argument. We have corrected this mistake by restoring postmortem damage information to Table 3, and we have added a MapDamage plot to the Supplementary Appendix (Appendix F, Figure F3).

We hope that you will find the revised manuscript greatly improved through the efforts of the editorial team and peer reviewers. Please do not hesitate to reach out to me with any questions you may have regarding the manuscript, and we look forward to sharing our work in *Nature:Human Behavior*.

Sincerely,



William Taylor, Ph.D.

Assistant Professor of Anthropology

Curator of Archaeology

University of Colorado-Boulder Museum of Natural History

Decision Letter, first revision:

Dear Dr Taylor,

RE: "Evidence for Early Dispersal of Domestic Sheep into Central Asia: genetic, archaeozoological, and biomolecular insights from Obishir V"

Thank you so much for submitting your revised manuscript and for all your work on the revision.

Thank you also for your email yesterday (15th Dec).

As you mentioned in your email, although your manuscript has been revised in response to reviewer comments, there remain some outstanding requests from our previous decision letter. Before we can send the manuscript back to our reviewers, we ask that you attend to the following:

- 1) Please provide a point-by-point response to each of our reviewers' comments (verbatim).
- 2) Please provide a version of the manuscript with tracked changes.

We make these requests in order to assist our reviewers in re-reviewing the manuscript. If you have any questions about either of these requests, please don't hesitate to contact me.

Please use the link below to submit your revised manuscript and related files:

[REDACTED]

Note: This URL links to your confidential home page and associated information about manuscripts you may have submitted, or that you are reviewing for us. If you wish to forward this email to co-authors, please delete the link to your homepage.

Thank you in advance for attending to these requests and I look forward to receiving your revised manuscript.

Sincerely,

Charlotte Payne
Editor
Nature Human Behaviour

Author Rebuttal, first revision:

16 December, 2020

Dear Dr. Payne,

Attached, please find a revised version of our manuscript, "Evidence for Early Dispersal of Domestic Sheep into Central Asia: genetic, archaeozoological, and biomolecular insights from Obishir V" for your consideration. The paper has been revised based on suggestions from three reviewers, who all found the manuscript to be well-written, with well-supported conclusions of broad significance for our understanding of human prehistory. In its revised form, we have addressed a number of minor issues with the text and supplementary material identified by the reviewers, and I hope you will find it to be a valuable contribution to *Nature: Human Behavior*.

Below, you will find a point-by-point description of the minor revisions we made in response to these reviewer comments, as well as a few minor points of clarification.

Reviewer 1

I recommend publication with very minimal edits. There are some small formatting issues that I'm sure will be resolved with typesetting (like centered figure captions).

In our revised manuscript, we have centered our figure captions.

Overall most of the manuscript was relatively concise and clear. I did find the use of "these two taxa" in the Discussion a little opaque; are the authors referring to wild or domestic forms when they say "a ratio of roughly 3:1 between these two taxa have characterized pastoralist archaeological assemblages in Central and Inner Asia since the Bronze Age"; I think that those few sentences could be clarified a bit.

Reviewer 1 identified ambiguity in reference to “these two taxa” in the first paragraph of the discussion, an issue that was also highlighted by Reviewer 2. As Reviewer 2 had correctly inferred, our intent here was to refer to the genus *Ovis* and *Capra*. In our revised manuscript, we have clarified the wording to this section to make this point clear.

Otherwise I think the amount of data provided in the supplemental information is appropriate and there is a good balance between what is presented and discussed in the main article and what is supplied for the reader. Though it is a small sample size, obviously the data are there to support the authors' argument that there were domestic sheep present at this early date, and that is a very important finding that should be published in this journal. I think the paper lays a foundation for future research in the region and makes a substantial contribution to our understanding of the spread of domesticates eastwards of the Fertile Crescent.

We thank the reviewer for this supportive and positive assessment of the manuscript.

Reviewer 2

The article "Evidence for Early Dispersal of Domestic Sheep into Central Asia: genetic, archaeozoological, and biomolecular insights from Obishir V" by Taylor et al. represents a scientific study of extreme importance in the field of zooarchaeology, covering the ancient genetics of domestic sheep, dispersal of domestic sheep after its initial domestication, and earliest insights into animal husbandry in Central Asia – topics that are long debated and are still debated by the scholars in different fields.

Although the study relies on rather scanty material from the site of Obishir V, Kyrgyzstan, the applied methods of interdisciplinary nature definitely provide a solid argument for a very important discovery: pastoral livestock species reached Central Asia millennia earlier as previously recognised, by around 6000 BCE. These results are a valuable input to the studies of domestication, dispersal of the domestic species from the domestication centre, and human/cultural migrations.

The article text is very well written and enjoyable to read. The figures, tables, and supplementary appendices illustrate and support the main text sufficiently. The selected methods have been well described, applied, and justified. Please note that the reviewer has no experience in thermoluminescence dating and cementum analysis and has somewhat experience in genetics, meaning that any potential shortages in these sections could have been missed.

We thank the reviewer for their thoughtful commentary and positive assessment of the manuscript.

There are only a few very minor remarks I would like to point out to the authors and when addressed, these would hopefully improve the quality of the article even further. These remarks are purely from the editorial point of view and do not diminish the quality of the article.

1. In Introduction, paragraphs 3 and 4, you have mentioned that “Domesticated sheep and goat are also thought to have been an important part of the subsistence economy at Djeitun (33 - 36).” and “Importantly, scholars have also argued that the Hissar economy included goat and sheep pastoralism (37).”. If possible, maybe you could add to those sentences, on which evidence these earlier suggestions rely – have the previous scholars relied here on the presence of ovicaprid remains?

In comments also echoed by Reviewer 3, Reviewer 2 requested clarification on the evidence used to infer the presence of domestic animals at Djeitun and in sites of the Hissar culture (Introduction, paragraphs 3 and 4). In our revised manuscript, we have added several sentences establishing that only indirect measures – a high frequency of *Ovis* and *Capra* remains that appear smaller than the region’s wild taxa – were used by earlier scholars to generate this hypothesis.

*2. In Discussion, paragraph 1, you have written that “Although wild *Ovis* and *Capra* in this region include the large argali sheep (*Ovis ammon*) and Siberian ibex (*Capra sibirica*), a ratio of roughly 3:1 between these two taxa have characterized pastoralist archaeological assemblages in Central and Inner Asia since the Bronze Age (22, 52), and are also typical of other early western Eurasian pastoral assemblages (53). Based on the results of ZooMS, these two taxa appear to comprise the majority of the unidentifiable fragments in the Layer 2 faunal assemblage.”. Please clarify, which “two taxa” you have meant in the last sentence? I believe it should be *Ovis* sp. and *Capra* sp.?*

This comment has been addressed by revisions in response to Reviewer 1, who identified the same issue.

3. In Discussion, under subchapter Evidence for pastoral management, paragraph 2, I believe there is something gone missing in the sentence “Our new findings indicate that the subsistence behaviors that underpinned these nascent interaction”

The second portion of this sentence appears to have been inadvertently deleted in our original submission. We have restored the complete sentence to the Discussion/”Evidence for pastoral management” subsection, paragraph 2

4. Few missing full stops and spaces noticed both in the main text and Supplementary Appendix.

We have carefully proofed our revised manuscript and the Supplementary Appendix for missing stops and spaces, and hope to have located and corrected those identified by Reviewer 2.

5. Regarding the figures, I have few comments:

a. Few of them appear to be of low resolution. However, if this is due to the submission process and better-quality figures will be uploaded after acceptance, please disregard this comment.

We believe that the poor resolution of figures was indeed linked to the submission process, and have ensured that our uploaded figures are of high resolution.

b. In Figure 3 (main text) and Figure C1 in Supplementary Appendix, the text is too small to follow – I would suggest enlarging it a bit.

For Figure 3 and Figure C1 in the Supplementary Appendix, we have increased the size of label text to increase legibility.

c. In Figure 3A (main text) I would like to see NISP values for each bar – it would make following the material details much easier.

We added NISP values to each bar in Figure 3A.

d. In Figure 3B (main text), I can see groupings for cervidae/ bovidae (4) and deer (5), but these are not listed in the text as the other groupings – maybe to list them in the text as well?

In Figure 3B, we have added text descriptions to the groupings for Cervidae/Bovidae and deer/saiga/sheep/gazelle, as requested by Reviewer 2.

e. Figure 5 is a single figure, but the in the main text it has been referred to as Figures 5A and 5B.

We have corrected references to Figure 5 in the main text, to correctly refer to Figure 5 and Supplementary Figure F1 (instead of 5A and 5B, as incorrectly noted in the first manuscript).

6. Regarding the Supplementary Appendix, I have few comments:

a. Consider improving the quality of Table B2.

Table B2, which in our initial submission was incorrectly included as an embedded image file rather than as text, has been revised to restore the text and thereby improve quality.

b. Large tables like Table C1, Table C2, Table F2, and Table F3 should be formatted for better readability or as written in the journal guidelines, “Tables larger than one page should be provided as tabular data files than a PDF.” In this case, just formatting them in the already existing Word file could work as well. Additionally, Table C2 is currently in both portrait and landscape.

Tables C1, C2, F2, and F3 have been formatted as tabular data, and included in a separate “Supplementary Data” file, correcting the formatting and readability issues identified by Reviewer 2. Appropriate references in the text have been updated to reflect this change.

c. In Table F1 caption, you mention numbering for sheep, but I can only see Capra in the table.

In our revision, our caption reference to sheep in Table F1 has been removed, to retain only information relevant to *Capra* sp.

Reviewer 3

This reviewer's comments were also largely positive, with the commenter highlighting the study's convincing and significant results. In addition to the comments addressed above, Reviewer 3 also requested a number of minor revisions which we have addressed here.

This study provides new and significant archeological evidence for an early spread of agropastoralism in Central Asia. Overall, this is a well written manuscript, presenting quite convincing results (although I mostly paid attention to the DNA part).

We thank the reviewer for this positive assessment of our submission.

Before diving into the detail of my review I would like to point out to the author and / or journal editor that there is no page or line number in this manuscript which is quite frustrating for reviewers.

In our revised manuscript, we have added page and line numbers to facilitate reading and review.

While I think the study is important, I found it hard to judge how novel the findings really are. I think this can be easily resolved as it stems from the fact that the introduction is lacking some important information. Here is a specific example: pg 4, first paragraph. We are given a lot of specific details about the evidence at Djeitun for crop farming but just a single sentence with no evidence about the sheep besides that they " are also thought to have been an important part... "? What is this based on? What is the evidence for sheep? Have these been dismissed? If yes, why? The same is true for the next paragraph, with the sentence: " Importantly, scholars have also argued that the Hissar economy included goat and sheep pastoralism (37). " - what evidence is presented in ref. 37? We are then told: " most of the relevant scholarly research on these cultures was performed prior to the widespread availability of scientific methods, including AMS radiocarbon dating. To date, no direct archaeobotanical or zooarchaeological evidence for food production has been found beyond the margins of the Kopet Dag piedmont of Turkmenistan prior to ca. 3500 BCE (22) ." which is hard to assess as we do not know what evidence past " scholarly research " was based on. This makes it difficult to assess whether this study is just confirming an hypothesis that was tested before, using new techniques, or whether the study is actually testing a long standing hypothesis that was never tested and whether the claims like " Until now, archaeological data suggested a relatively late arrival of the agropastoral economy to interior Central Asia, around 3500 BCE ." is indeed a valid conclusion.

These comments are similar to observations made by other reviewers, and have been addressed through clarification and additions to this section of the revised manuscript. Our revised introduction now makes it clear that the suggestion of domestic sheep and goat at Djeitun has not been rejected, but neither has it been verified through scientific methods. We have also updated a sentence in Paragraph 5 to clarify that no similar arguments have been made for the Ferghana region, which lies nearly 1400 km to the northeast of Djeitun. The revised language and new sentences in this section now make it clear that our study, while contextually supported by earlier work in Turkmenistan and Tajikistan hypothesizing an early eastward dispersal of Neolithic economies, is testing and confirming a new, previously untested hypothesis.

The mitochondrial DNA analysis is mostly fine (although very brief) for the simple purpose of validating whether the specimens are domestic or wild. I am surprised that the author did use Maximum Parsimony though - is there any reason for this?

Reviewer 3 found our mtDNA analysis to be brief but well-supported, but inquired as to the reasoning behind our selection of Maximum Parsimony over other alternatives (such as Maximum Likelihood with bootstrapping, neighbor-joining with bootstrapping, or Bayesian methods). We respond that while Reviewer 3 is generally correct, Maximum Parsimony still has currency in some areas of ancient genomics, as it minimizes the number of evolutionary events needed to explain the data. In comparison, Maximum Likelihood and Bayesian inference methods incorporate an evolutionary model and prior knowledge about the data, respectively, that may not be as suited to all circumstances.

Although I doubt it would change the result, it is much better practice to use more up to date methods, such as Maximum Likelihood (with bootstrapping), Bayesian methods, or even Neighbor joining (also with bootstrapping).

In response to Reviewer 3's suggestion, we have repeated our analyses using a Maximum Likelihood approach with bootstrapping, and added this information to Supplementary Appendix F. The Maximum Likelihood tree supports the typology of the Maximum Parsimony tree proposed in the main text.

I am a bit puzzled by the PCA on nuclear data - the authors say that their samples :'' fell within the variation of Ovis aries and were confirmed to be sheep '' - but these were projected right onto a PCA with no wild sheep right? Maybe I am missing something here but how can the authors, without wild sheep in this analysis, conclude that these specimens are domestic sheep and not wild. I suggest the author run an analysis that actually includes wild sheep data to test their hypothesis or remove this analysis from the paper. Although I think this is an issue, it does not affect the overall conclusion of the paper as mitochondrial analysis is enough to conclude that their specimens are indeed domestic sheep.

In our initial submission, we based our assertion that Obishir sheep fell within the variation of *O. aries* on our phylogenetic analyses, which included important wild sheep taxa including the argali sheep, *O. ammon*, the Urial sheep, *O. vignei*, and the Cypriot mouflon, *O. orientalis ophion*. However, we recognize the added value of incorporating wild sheep data into our subsequent Principal Components Analysis. While whole-genome data for wild sheep taxa are less widely available than the mtDNA sequences used in our phylogeny, we were able to isolate a number of Asiatic mouflon genotypes – the most closely related wild sheep taxa to the domestic *O. aries* – from a recently published study (<https://www.nature.com/articles/s41467-020-16485-1>). In our revised analysis, we have incorporated these wild sheep genomes into our PCA along with two other wild taxa (*O. canadensis* and *O. dalli*), which support our interpretation that the specimens are indeed domestic sheep.

Minor comments:

" We observed high variability in endogenous DNA preservation across the fragments, ranging from 0.2% to 8.3%, and all five specimens exhibited patterns of post- mortem DNA damage consistent with ancient DNA (Table 3) ." It seems that Table 3 might be miss cited here - which information in this table relates to damage? I would include a mapdamage plot in the supplementary for this which would be much more convincing.

We have corrected this mistake by restoring postmortem damage information to Table 3, and we have added a MapDamage plot to the Supplementary Appendix (Appendix F, Figure F3).

Figure 6 - I would include some information about how this tree was built. Although it is probably fine, Maximum parsimony is a bit outdated - I would build a maximum likelihood or a bayesian tree to make it more convincing.

We have added this information about how Figure 6 was built to Supplementary Appendix F, supported with a note to this effect to the caption for Figure 6 in the main text. As noted above, we have also augmented our Maximum Parsimony tree with additional Maximum Likelihood analysis added to the supplementary appendix.

In addition to these reviewer comments, we have redone our chronological modeling using the newly released INTCAL20 calibration curve, which was not available at the time of initial submission. We hope that you will find the revised manuscript greatly improved through the efforts of the editorial team and peer reviewers. Please do not hesitate to reach out to me with any questions you may have regarding the manuscript, and we look forward to sharing our work in *Nature:Human Behavior*.

Sincerely,



William Taylor, Ph.D.

Assistant Professor of Anthropology

Curator of Archaeology

University of Colorado-Boulder Museum of Natural History

Decision Letter, second revision:
--

18th January 2021

*Please ensure you delete the link to your author homepage in this e-mail if you wish to forward it to your co-authors.

Dear Dr Taylor,

Thank you once again for submitting your revised manuscript, entitled "Evidence for Early Dispersal of Domestic Sheep into Central Asia: genetic, archaeozoological, and biomolecular insights from Obishir V," and for your patience during the re-review process.

Your manuscript has now been evaluated by our referees, and in the light of their advice I am delighted to say that we can in principle offer to publish it. First, however, we would like you to revise your paper to address the points made by Reviewer 2, and to ensure that it complies with our Guide to Authors at <http://www.nature.com/nathumbehav/info/gta>.

One of the main reasons for delays in formal acceptance is failure to fully comply with editorial policies and formatting requirements. To assist you with finalizing your manuscript for publication, I attach a checklist that lists all of our editorial policies and formatting requirements. I also attach a template document, which exemplifies our policies and formatting requirements.

Please attend to *every item* in the checklist and upload a copy of the completed checklist with your submission. I have highlighted in the checklist items that require your attention. I also mention here a few points that are frequently missed and can cause delays:

- 1) Ensure that all corresponding authors have linked their ORCID to their account on our online manuscript handling system. This is very frequently missed and invariably causes delays in formal acceptance.
- 2) Ensure that you provide all of the materials requested in the attached checklist and below with your final submission.

Nature Human Behaviour offers a transparent peer review option for new original research manuscripts submitted from 1st December 2019. We encourage increased transparency in peer review by publishing the reviewer comments, author rebuttal letters and editorial decision letters if the authors agree. Such peer review material is made available as a supplementary peer review file. **Please state in the cover letter 'I wish to participate in transparent peer review' if you want to opt in, or 'I do not wish to participate in transparent peer review' if you don't.** Failure to state your preference will result in delays in accepting your manuscript for publication.

Please note: we allow redactions to authors' rebuttal and reviewer comments in the interest of confidentiality. If you are concerned about the release of confidential data, please let us know specifically what information you would like to have removed. Please note that we cannot incorporate redactions for any other reasons. Reviewer names will be published in the peer review files if the reviewer signed the comments to authors, or if reviewers explicitly agree to release their name. For more information, please refer to our [FAQ page](https://www.nature.com/documents/nr-transparent-peer-review.pdf).

We hope to hear from you within 10 days; please let us know if the revision process is likely to take longer.

To submit your revised manuscript, you will need to provide the following:

- Cover letter
- Point-by-point response to the reviewers (if applicable)
- Manuscript text (not including the figures) in .docx or .tex format
- Individual figure files (one figure per file)
- Extended Data & Supplementary Information, as instructed
- Reporting summary

- Editorial policy checklist
- Third-party rights table (if applicable)
- Suggestions for cover illustrations (if desired)

Consortia authorship:

For papers containing one or more consortia, all members of the consortium who contributed to the paper must be listed in the paper (i.e., print/online PDF). If necessary, individual authors can be listed in both the main author list and as a member of a consortium listed at the end of the paper. When submitting your revised manuscript via the online submission system, the consortium name should be entered as an author, together with the contact details of a nominated consortium representative. See <https://www.nature.com/authors/policies/authorship.html> for our authorship policy and <https://www.nature.com/documents/nr-consortia-formatting.pdf> for further consortia formatting guidelines, which should be adhered to prior to acceptance.

Reviewer Recognition:

In recognition of the time and expertise our reviewers provide to Nature Human Behaviour's editorial process, we would like to formally acknowledge their contribution to the external peer review of your manuscript entitled "Evidence for Early Dispersal of Domestic Sheep into Central Asia: genetic, archaeozoological, and biomolecular insights from Obishir V". For those reviewers who give their assent, we will be publishing their names alongside the published article.

Forms:

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If you have any further questions, please feel free to contact me.

With best regards,

Charlotte Payne
Editor
Nature Human Behaviour

Reviewer #1:

Remarks to the Author:

After reading the authors' response to reviewers and the revised manuscript, I would recommend this article for publication without further revision. The concerns voiced by all three reviewers have been addressed and the manuscript, which was already high quality, has been improved by these edits.

Reviewer #2:

Remarks to the Author:

Dear authors,

Thank you for a detailed response. You have addressed all my previous comments sufficiently and I agree with the revision. In my opinion, this article is now ready to proceed further in the publication process.

Tiny typos that caught my eye when browsing through the revised manuscript (clean version):

- 1) line 161 - some typos;
- 2) Figure 2 caption - it seems that some editing is needed in presenting the layer numbers;
- 3) Figure 5 caption - A and B are missing, although they appear in the main text as Figure 5A and Figure 5B.

Good luck!

Reviewer #3:

Remarks to the Author:

The authors have constructively addressed all my concerns - this is solid piece of work which, as far as I am concerned, can be published as is.

Final Decision Letter:

Dear Dr Taylor,

We are pleased to inform you that your Article "Evidence for Early Dispersal of Domestic Sheep into Central Asia", has now been accepted for publication in Nature Human Behaviour.

Before your manuscript is typeset, we will edit the text to ensure it is intelligible to our wide readership and conforms to house style. We look particularly carefully at the titles of all papers to ensure that they are relatively brief and understandable.

Once your manuscript is typeset and you have completed the appropriate grant of rights, you will receive a link to your electronic proof via email with a request to make any corrections within 48 hours. If, when you receive your proof, you cannot meet this deadline, please inform us at rjsproduction@springernature.com immediately. Once your paper has been scheduled for online publication, the Nature press office will be in touch to confirm the details.

Acceptance of your manuscript is conditional on all authors' agreement with our publication policies (see <http://www.nature.com/nathumbehav/info/gta>). In particular your manuscript must not be published elsewhere and there must be no announcement of the work to any media outlet until the publication date (the day on which it is uploaded onto our web site).

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We look forward to publishing your paper.

With best regards,

Charlotte Payne
Editor
Nature Human Behaviour